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Challenges to evolution as a core concept in college biology: Silence on religion and conflicting goals for acceptance

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Abstract

National recommendations, including the *Vision and Change* report, have proposed that evolution be taught as a core concept in biology, but do college biology instructors teach it as such? We surveyed 217 instructors across the United States and most recognized evolution as a core concept but reported teaching at least one biology course in which they did not mention evolution or only included it in a few lessons, suggesting they are not following recommendations to integrate core concepts throughout courses. Instructors were split on whether a goal of evolution instruction was for acceptance or solely for understanding, and most instructors reported not discussing potential compatibility between religion and evolution. Further, half of instructors reported they did not think a student fully accepted evolution if they thought a higher power created evolution. We discuss the implications of these results for realizing evolution as a core concept in college biology instruction.

Keywords **evolution, college, science and religion, nature of science, higher education**

Vision and Change recommends teaching evolution as a core concept throughout all undergraduate biology courses

The *Vision and Change* report was a call to action for biologists to transform undergraduate biology education to reflect rapid advances in biology and in our understanding of how students learn (American Association for the Advancement of Science 2011). This initiative started in 2006 with a National Science Foundation working group that led to a series of regional meetings and then, in 2009, culminated with a national meeting of over 500 biologists and biology educators. The summary report, titled *Vision and Change*, was published in 2011.

A major outcome was the identification of five core concepts essential for students' biological literacy (table 1). Importantly, these concepts were recommended to be "woven into the curriculum and reinforced throughout all undergraduate biology coursework" so that all students, including majors and nonmajors, would engage with them repeatedly, using them as an organizing framework for thinking about biology (American Association for the Advancement of Science 2011, Brownell et al. 2014). Evolution, one of these five core concepts,

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Table 1. Core concepts of biology that *Vision and Change* recommends to be woven throughout undergraduate biology courses (American Association for the Advancement of Science 2011).

Core concept	Definition
Evolution	*The diversity of life evolved over time by processes of mutation, selection, and genetic change.*
Structure and function	*Basic units of structure define the function of all living things.*
Information flow, exchange, and storage	*The growth and behavior of organisms are activated through the expression of genetic information in context.*
Pathways and transformations of energy and matter	*Biological systems grow and change by processes based upon chemical transformation pathways and are governed by the laws of thermodynamics*
Systems	*Living systems are interconnected and interacting.*

is described as "the fundamental organizing principle over the entire range of biological phenomena," a sentiment echoed in other recommendations (NGSS Lead States 2013, Brownell et al. 2014). There have been efforts to collect evidence on general awareness and understanding of *Vision and Change* through quantifying frequency of citations in NSF grant proposals (Vasaly et al. 2014), targeted interviews with instructors (Hsu et al. 2025), and assessing departmental change through the NSF Partnership for Undergraduate Life Sciences Education (PULSE) program (Peteroy-Kelly et al. 2019). However, we do not know how broadly *Vision and Change* recommendations have been implemented in undergraduate biology instruction. Here, we were interested in gathering national data on how college biology instructors have implemented one specific recommendation of *Vision and Change*: teaching evolution as a core concept.

Evolution is often rejected by students, but they will need to see it as valid if they are going to use it as a foundation for their biology knowledge

Distinct from the other core concepts of *Vision and Change*, there is societal controversy surrounding evolution within the United States, which requires US-based undergraduate biology instructors to not only contemplate their students' understanding of this core concept, but also acceptance. Although nearly all scientists agree that humans evolved from prior species (Masci 2019), there is persistent skepticism about human evolution both in the US general public and among undergraduate biology students. Studies over the past 50 years have shown that around half of the US public does not accept human evolution from nonhuman species (Gallup 2024), with those in the United States being significantly more likely to reject human evolution than in many other national contexts (Miller et al. 2006, Riley and Elsdon-Baker 2025). A study of almost two thousand undergraduate biology students in the United States documented that about 30% do not accept that all of life shares a common ancestor (Barnes et al. 2020a). Most topics in biology are taught with student understanding in mind, and it is assumed that acceptance will follow from gaining sufficient understanding (e.g., photosynthesis). Thus, it is not surprising that some instructors have reported that they think if students understand the theory of evolution, then they will accept the fundamental role of evolution in biology. In other words, some instructors do not see a difference between teaching toward the understanding of evolution versus the acceptance of evolution (Barnes and Brownell 2016). However, students have reported they "just learn [evolution] and memorize it for exam purposes" but that they "don't necessarily believe it" (Barnes et al. 2017b). So, understanding evolution does not necessarily lead to acceptance (Brem et al. 2003, Sinatra et al. 2003, Nadelson and Sinatra 2010, Hermann 2012). If students understand evolution but do not accept it as a real phenomenon, they are unlikely to see it as the foundation from which to build their biology knowledge. Thus, unlike other core concepts, if instructors want students to incorporate evolution into their construction of biological knowledge and use it as a tool for solving biology problems in the real world, they may need to increase students' understanding *and* acceptance of evolution. But do instructors teach evolution in a way that leads to acceptance, or do they even consider acceptance as a goal of their instruction? In an interview study with 32 college biology instructors, most instructors said they only aim for understanding of evolution because they felt that a goal of acceptance would be manipulative or authoritarian (Barnes and Brownell 2016), but we know of no research that has explored how common these viewpoints are among college biology instructors.

What is evolution acceptance and what affects it?

Evolution acceptance has been defined in several different ways. Evolution instructors have reported in interviews that in order to accept evolution, a student must accept that all life shares a common ancestor and that natural selection is responsible for the diversity of life on Earth (Barnes and Brownell 2016). Recently, a consensus definition of evolution acceptance was agreed upon and published by a large interdisciplinary group of biologists and evolution education researchers. They concluded that evolution acceptance is: "agreeing that evolution is valid and the best explanation from science for the unity and diversity of life on Earth, which includes speciation, the common ancestry of life, and that humans evolved from non-human ancestors" (Barnes et al. 2024), which is the definition we subscribe to in this work.

What affects student acceptance of evolution? First, the context of evolution matters. Students rarely reject ideas such as natural selection leading to small changes in species, but more often reject human evolution, specifically that all of life, including humans, shares a common ancestor (Nadelson and Southerland 2010, 2012, Barnes et al. 2019, Misheva et al. 2023, Aini et al. 2024a). Further, the extent to which students understand evolution and the nature of science *is* positively related to acceptance (e.g., Rutledge and Warden 2000, Ha et al. 2012, Rissler et al. 2014, Glaze and Goldston 2015, Glaze et al. 2015, Dunk et al. 2017, Barnes et al. 2019, Fiedler et al. 2019, Jensen et al. 2019, Betti et al. 2020, Fahmy et al. 2025). However, the relationship between understanding and acceptance is strongly shaped by religious identity. For example, a study of thousands of undergraduate biology students found that for nonreligious students a higher score on an evolution test strongly correlated with higher acceptance that

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life shares a common ancestor. However, religious students with high or low evolution understanding were equally likely to reject the common ancestry of life (Aini et al. 2024a). This implies that increasing understanding of evolution may only increase acceptance among people who do not perceive a conflict with their religious identity. In fact, a large-scale study on factors affecting evolution acceptance showed that *perceived conflict* between religion and evolution predicts evolution acceptance far more than understanding of evolution (Barnes et al. 2021).

Do students have to reject belief in God/god(s) to accept evolution?

There *is* conflict between certain religious beliefs and evolution, particularly a literal interpretation of some religious texts, but there are also many examples of religious organizations and religious individuals who have been able to reconcile religious beliefs with evolution so that they believe and advocate for potential compatibility between religion and evolution (The Clergy Letter Project 2004, Tharoor 2014, BioLogos 2023, The Society of Catholic Scientists). National context also matters; for example, religious individuals in the United Kingdom perceive less conflict between their religious beliefs and evolution than religious individuals in the United States (Riley and Elsdon-Baker 2025). Additionally, there are scientific organizations and scientists, both religious and nonreligious, who are proponents of reducing perceived conflict between religion and evolution because the bounded nature of science asserts that religion and science answer different questions. Although they can inform each other, questions about the existence of a higher power are outside the purview of the natural world and what science can study (Gould 1997, AAAS 2006, Collins 2007, Miller 2007, DoSER 2020). Thus, the existence of God/god(s) and their influence on evolution are outside the scope of science, making science and evolution agnostic (i.e., not making claims about the existence or influence of supernatural entities) rather than atheistic (i.e., excluding the possibility of the existence or influence of supernatural entities). Indeed, the word "agnostic" was coined by evolutionary biologist Thomas Henry Huxley to specifically describe science and settle debates between science and religion (Huxley and Huxley 1900). In this study, we define evolutionary science as agnostic about the existence or influence of supernatural entities on the natural world. But do college biology instructors think students have to reject a belief that God/god(s) created evolution for them to fully accept it?

Evolution instructors may think students need to adopt an atheistic definition of evolution to accept it, which would likely increase perceived conflict between religion and evolution and decrease acceptance (Barnes et al. 2020a, Edwards et al. 2025c). In interviews, many instructors reported that a student could not believe that a God/god(s) "planned, started, or guided evolution" if they were to fully accept evolution (Barnes and Brownell 2016). Many of the biology instructors also said they completely avoid the topic of religion when teaching evolution, which leaves little room for them to discuss the potential compatibility between the two (Barnes and Brownell 2016) even though showing this compatibility can be effective for increasing acceptance (Winslow et al. 2011, Lindsay et al. 2019, Green and Delgado 2021). There were even some instructors who reported being outwardly antagonistic towards religion when teaching evolution. However, we do not know the prevalence of these attitudes and behaviors among college biology instructors. If we want students to accept evolution and use it as a core concept in their thinking of biology, it is important to understand the extent to which instructors address potential compatibility between evolution and religion when teaching evolution and if they believe students need to adopt atheistic definitions of evolution for them to accept evolution.

The current study

Our goal for this current study was to survey a national population of biology instructors at secular institutions of higher education and examine what proportion of instructors (1) are aware of and teach evolution as a core concept in their biology content courses, (2) see acceptance of evolution as a goal of their instruction, (3) hold agnostic or atheistic definitions of evolution acceptance, and (4) address the relationship between evolution and religion as compatible or adversarial. Further, we also examined whether the demographic variables of gender and religious status of the instructor were predictive of these outcome variables.

Our author team has a variety of positionalities that have informed our interest in this project that can be found in the supplemental material.

Methods

This study was done in accordance with an approved IRB from Arizona State University (#00021954).

Recruitment

In an effort to reduce self-selection bias, we collected instructor names and emails from every biology instructor in the United States at a secular institution of higher education from their institutional websites and sent recruitment emails to those instructors. To further reduce response bias, the recruitment email invited recipients to complete a survey about their experience teaching undergraduate biology but intentionally did not mention *Vision and Change*, evolution, evolution acceptance, or religion. Instructors were incentivized to complete the survey in two ways: the first 50 respondents received a gift card for $100, and all respondents were entered into a raffle for one of two $500 gift cards.

We sent emails using a mail-merge system to personalize the emails with their names in Spring 2025. Out of 275 responses, 217 instructors fulfilled the criteria of having taught a content-focused undergraduate biology course and were included in our study. It is impossible to ascertain how many people received and read the email, so we are unable to calculate a response rate.

Survey design

We designed a closed-ended survey to answer our research questions. To gauge instructor awareness about the *Vision and Change* report, we created a binary closed-ended question with follow-up questions about instructors' awareness of recommendations for teaching each of the core concepts of *Vision and Change*. Regardless of the instructor's stated knowledge of *Vision and Change*, all respondents were provided definitions of the core concepts and, for each core concept, asked if they used that concept to guide their teaching.

To ascertain the extent to which instructors were incorporating the concept of evolution into each of their courses, we asked all respondents to select the statement that best reflected how evolution was

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taught in their course: the main focus of the course, a theme weaved throughout the entire course, only the focus of several lessons over multiple class periods, only the focus of a single class period, or only mentioned or used as an example or in an exercise. To explore instructor goals while teaching evolution, we asked each instructor to select whether their goal was for students to understand evolution only, accept evolution only, or understand and accept evolution.

As a proxy to assess whether instructors held atheistic or agnostic definitions of evolution acceptance, we asked all respondents to consider a scenario about a student in their class who thinks that (1) all of life on earth shares a common ancestor, (2) natural selection, genetic drift, gene flow, and mutation have been responsible for the diversification of life, and (3) a higher power purposefully created this process. Instructors were asked if, in their opinion, the student fully, partially, or does not accept evolution. For analyses, we collapsed the responses "does not accept evolution" and "partially accepts evolution" into the category of "atheistic definition of evolution acceptance" because both responses indicate that the instructor thinks that full acceptance of evolution excludes the possibility of God/god(s) involvement. The response "fully accepts evolution" was reclassified as "agnostic definition of evolution acceptance" to reflect that these instructors think that students can maintain a belief in God/god(s) and accept evolution fully.

To determine if and how instructors were addressing religion in classes where they reported teaching evolution, we asked respondents to select one of the following choices: I do not talk about the relationship between religion and evolution, I present religion and evolution as mostly adversarial, I present religion and evolution as mostly compatible, or I present religion and evolution as both compatible and adversarial in about equal amounts.

Survey validity evidence

To support the validity of our survey instrument, we conducted think-aloud response process interviews (AERA et al. 2014) with five biology instructors. The survey items were revised following each interview based on comments or questions from interview participants. The survey was additionally reviewed by four members of the research team for clarity and revised before finalizing.

Survey analysis

Demographic information was collected to determine how representative our sample was and to control for gender and religious status in our regression analyses (see supplemental material for the specific demographic data collected and how decisions were made to collapse categories).

All statistical analyses were conducted using R v. 4.4.1 (R Core Team 2024).

Regressions

To determine the relationships between demographic variables and different evolution teaching variables, we used binary logistic regression, where the outcome variables were coded as 1 or 0. We used the glm() function in the base R package "stats" to fit our models using the "binomial" family. We report coefficients and statistical significance when a variable reaches significance at $p < .05$. Full regression tables for all analyses reported in this manuscript, with coefficients, confidence intervals, and *p*-values, can be found in the supplemental material.

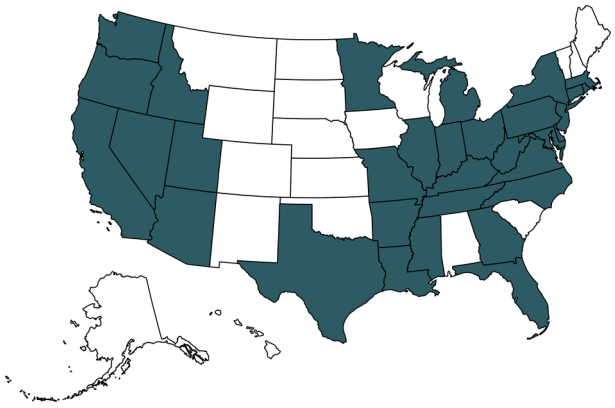


Figure 1. A map of the distribution of study participants.

Co-linearity tests

To test for multicollinearity of variables in our models, we used the vif() function of the R package "car" (Fox and Weisberg 2019) to calculate the Variance Inflation Factor (VIF). A VIF below 5 is considered acceptable multicollinearity (Menard 2001). All of our final models showed a VIF of less than 5 for each variable included.

Chi-squared and Fisher's exact tests

Due to an insufficient sample size of instructors who both (1) reported discussing religion and (2) reported teaching for understanding, not acceptance, we performed a chi-squared test to investigate the relationship between evolution instructors' goals in teaching evolution (i.e., teaching for understanding or teaching for both understanding and acceptance) and whether instructors reported explicitly discussing the relationship between religion and evolution in class, rather than perform a regression analysis.

We also expected that there may be relationships between religious affiliation and gender, given that, overall, women tend to be more religious than men (Pew Research Center 2016) and that instructors' perceptions of evolution acceptance as atheistic or agnostic would vary across religious identities, but we did not have the sample size to include interactions between these factors in our regressions, so we also investigated this relationship using chi-squared tests (see supplemental material)

Pearson's chi-squared tests and Fisher's exact tests were conducted using the chisq.test() and fisher.test() functions in the base R package "stats."

Results

Study population

A total of 217 biology instructors reported teaching 488 unique courses at 87 different institutions across 32 states (figure 1). Of these courses, 33 were solely focused on evolution. Although we had representation across institution types, instructors in our sample were primarily from research institutions. When asked to select one or more of the biology subdisciplines that they identified with, instructors fairly evenly identified among the three typical subdisciplines of

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biology, with 39.2% identifying as cell/molecular biology, 35.9% identifying as physiology/organismal biology, and 46.1% identifying as ecology/evolutionary biology. Notably, 11.4% of instructors identified as discipline-based education researchers, which is a population of instructors who we would predict would be more familiar with *Vision and Change*. Instructors in our sample were primarily white, nonreligious, and held the title of professors, with a fairly even split between men and women. See table 2 for demographic information about our sample and figure 1 for a map of representation across states.

Finding 1: Only about half of undergraduate biology instructors sampled were aware of the recommendations of *Vision and Change* to teach evolution as a core concept, and even though 87% reported using evolution as a core concept, fewer than 40% reported using it as a core concept in all their courses

A total of 55.7% (121/217) of instructors were aware of the *Vision and Change* initiative. Of the instructors who were aware of *Vision and Change*, 95% (115/121) were also aware that evolution is a core concept of *Vision and Change*.

Since some instructors may not be familiar with *Vision and Change*, but still consider evolution to be a core principle or idea of biology (Alles 2001, Michael et al. 2009, Wood 2009, Scheiner 2010, Quinn et al. 2011), we asked *all* instructors in our sample about their use of evolution as a core concept. Out of all instructors in our sample, 87.6% (190/217) reported using evolution as a core concept to guide their teaching.

To determine if evolution was being used as a core concept throughout the biology curriculum, we focused on instructors teaching the biology content courses that were not solely focused on evolution. In these courses, instructors reported using evolution as a core concept to variable extents. Of 217 instructors, only half (51.6%, 112/217) reported teaching a biology content course in which they integrated evolution as a core concept throughout the course. Notably, about one-fourth of instructors (22.6%, 49/217) reported not including evolution *at all* in at least one of their courses. Many instructors taught courses in which they did include evolution instruction, but not as a core concept. For instance, 12.4% (27/217) of instructors only mentioned evolution as an example in at least one of their courses, and 25.3% (55/217) reported only teaching evolution as the focus of one or a few class sessions in at least one of their courses. Notably, only one-third of instructors (33%, 71/217) used evolution as a core concept woven throughout *all* of their courses.

Instructor familiarity with evolution as a core concept of *Vision and Change* was not associated with differences in whether they integrated evolution as a core concept throughout at least one course. Likewise, no differences were observed based on gender or religious affiliation or religious background (see supplemental material for full regression tables).

Finding 2: Almost half of biology instructors sampled perceived that a student cannot fully accept evolution if they also believe that a higher power had a role in creating the process of evolution

We presented instructors with a possible scenario of student beliefs about evolution and a higher power, specifically that the student thinks evolutionary processes drive diversification of species and that all of life shares common ancestry but believes that a higher power created evolution. Although just over half of instructors (51.6%, 111/215) thought this student fully accepted evolution, the remaining instructors (48.4%, 104/215) did not agree that this student fully accepted evolution. Most of these instructors thought the student *partially* accepted evolution, but a small percent (6%, 13/215) did not think the student accepted evolution at all (figure 2).

When we examined whether religious status or gender affected this perception, the only significant difference observed was gender in that women were more likely to report the student in the scenario as fully accepting evolution (0.76832, $p = .00829$).

Finding 3: Biology instructors are split over whether acceptance of evolution is a goal of their instruction

Almost half of instructors (44.7%, 96/215) report that their goal is for students to understand and accept evolution, whereas over half (55.3%, 119/215) report that their goal is student understanding of evolution alone (figure 3).

When we examined whether gender, religious status, or holding an atheistic definition of evolution acceptance predicted whether instructors had a goal for understanding alone or understanding and acceptance, we found significant effects of all three. Specifically, women were more likely than men to report teaching for *understanding only* (-0.8044, $p = .00915$). In contrast, instructors who held an agnostic definition of evolution were more likely to report teaching for *acceptance and understanding* compared to instructors who held an atheistic definition (0.6731, $p = .02980$). The regression showed that nonreligious faculty, both those who were formerly religious and those who were never religious, were significantly more likely to report a goal of understanding and acceptance than their Christian peers. The difference between nonreligious and Christian faculty was larger for the nonreligious faculty who were formerly religious (1.0506, $p = .01425$) than for those who had never been religious (0.8059, $p = .04389$).

Finding 4: Most instructors do not talk about the relationship between religion and evolution in their courses

When we asked instructors who reported teaching about evolution whether they engaged in any discussion about religion and evolution, the majority of instructors reported that they do not discuss the relationship between evolution and religion (70.2%, 120/171). Some instructors who discussed religion and evolution reported presenting them as adversarial (12.3%, 21/171). Less than one-fifth of instructors (17.5%, 30/171) discussed potential compatibility between evolution and religion (figure 4). For the instructors who taught courses solely on evolution, within these courses, 27.5% (8/33) did not discuss religion, 37.9% (11/33) discussed it as compatible with evolution, and 34.5% (10/33) discussed it as adversarial to evolution.

We also investigated whether an instructor's goal for evolution education (teaching for understanding only or understanding and acceptance), their definition of evolution acceptance (atheistic versus agnostic), their religion, or their gender were associated with discussing religion when teaching evolution. We conducted three separate chi-squared tests and one Fisher's exact test because we did not have the sample size for regressions. Instructors with a goal of acceptance were more likely to discuss evolution and religion rather than avoid the topic ($x^2 = 7.67$, $p = .0056$) (figure 5). Definition, religion, and gender were not significant (see supplement material for results of nonsignificant tests).

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Table 2. Study participant demographics.

	n (%)		n (%)
Gender		**Race/ethnicity**	
Man	108 (49.8%)	Asian	14 (6.5%)
Woman	101 (46.5%)	Black	4 (1.8%)
Non-binary/Other gender	4 (1.8%)	Hispanic	12 (5.5%)
No answer	4 (1.8%)	White	164 (75.6%)
Religion		Other race/ethnicity	14 (6.5%)
Christian	51 (23.5%)	No answer	9 (4.1%)
Other religion	27 (12.4%)	**Carnegie Classification**	
No religion	128 (59.0%)	R1	131 (60.4%)
Formerly religious	*54 (42.2%)*	R2	21 (9.7%)
Never religious	*73 (57.0%)*	Doctoral granting	4 (1.8%)
No answer	*1 (0.8%%)*	Master's granting	11 (5.1%)
No answer	11 (5.1%)	Primarily undergraduate	26 (12.0%)
Position type		Community college	13 (6.0%)
Adjunct instructor	10 (4.6%)	No answer	11 (5.1%)
Full-time instructor	24 (11.1%)		
Assistant Professor	18 (8.3%)		
Associate Professor	63 (29.0%)		
Full Professor	87 (40.1%)		
Other position	15 (6.9%)		
Sub-discipline (multiselect)			
Anatomy & Physiology	33 (15.2%)		
Cell/Molecular	85 (39.2%)		
Ecology/Evolution	100 (46.1%)		
Discipline-Based Education	24 (11.1%)		
Organismal	45 (20.7%)		
Other	34 (15.7%)		

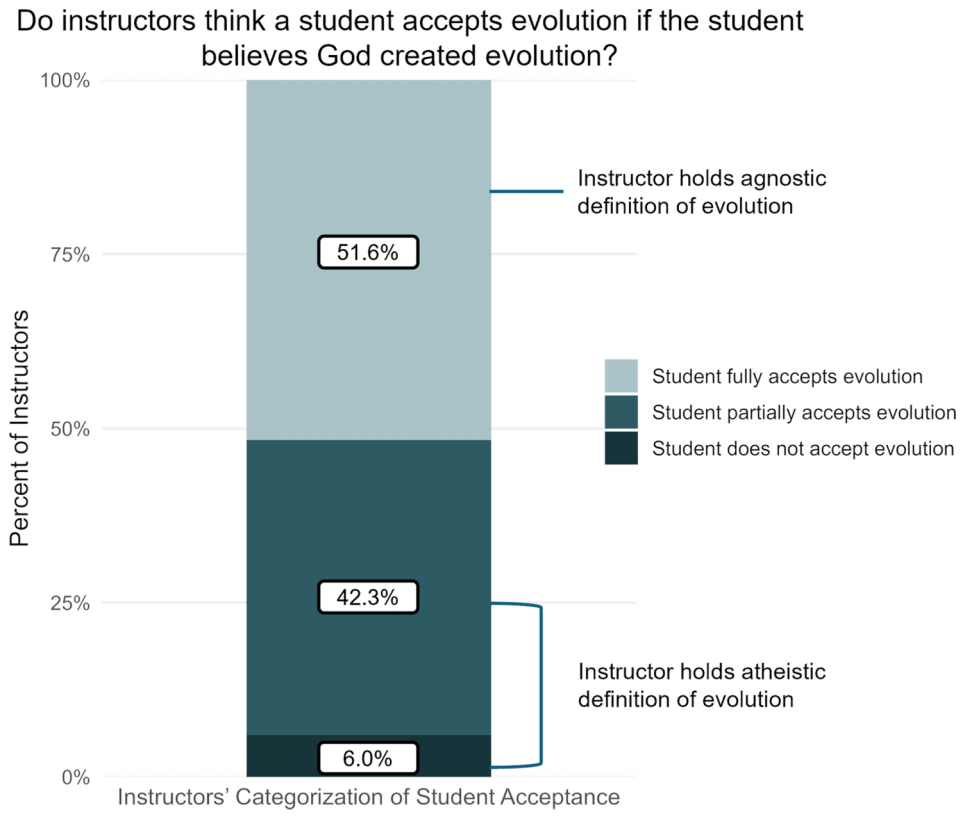


Figure 2. Instructor responses ($n = 215$) to a question presenting a scenario of a student who thinks evolutionary processes drive diversification of species, that all life shares a common ancestor, and that a higher power had a role in creating evolutionary processes. Instructors who did not select that the student fully accepts evolution were labeled as holding an atheistic definition of evolution because this indicates the instructor believes evolutionary theory makes claims about the influence of God/god(s).

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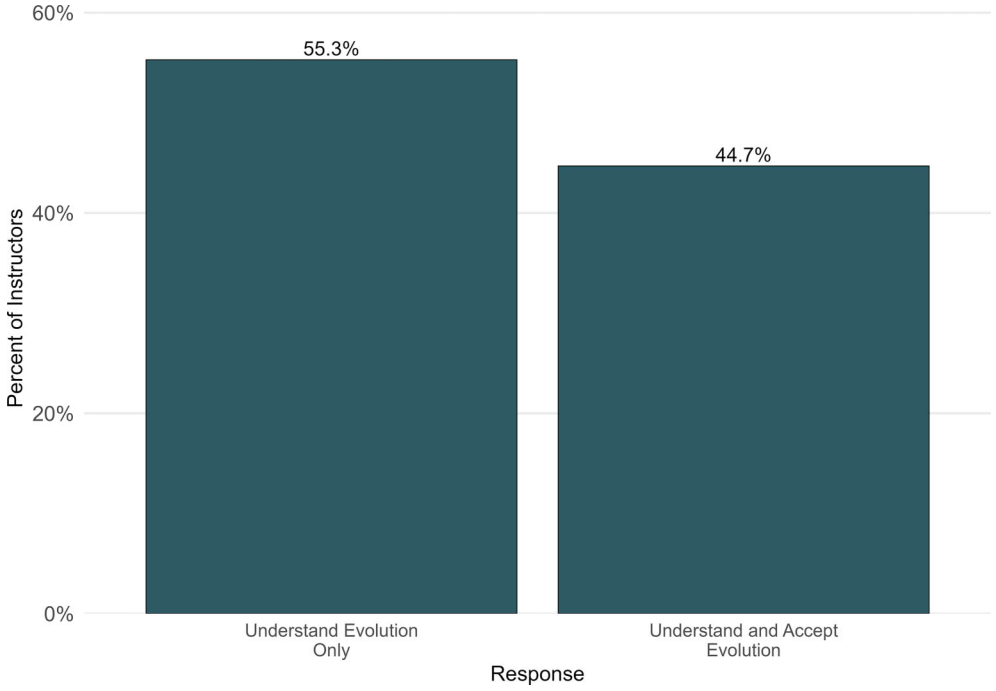


Figure 3. Instructors' stated goals for students during instruction on evolution ($n = 215$). The majority of faculty teach for understanding compared to understanding and acceptance.

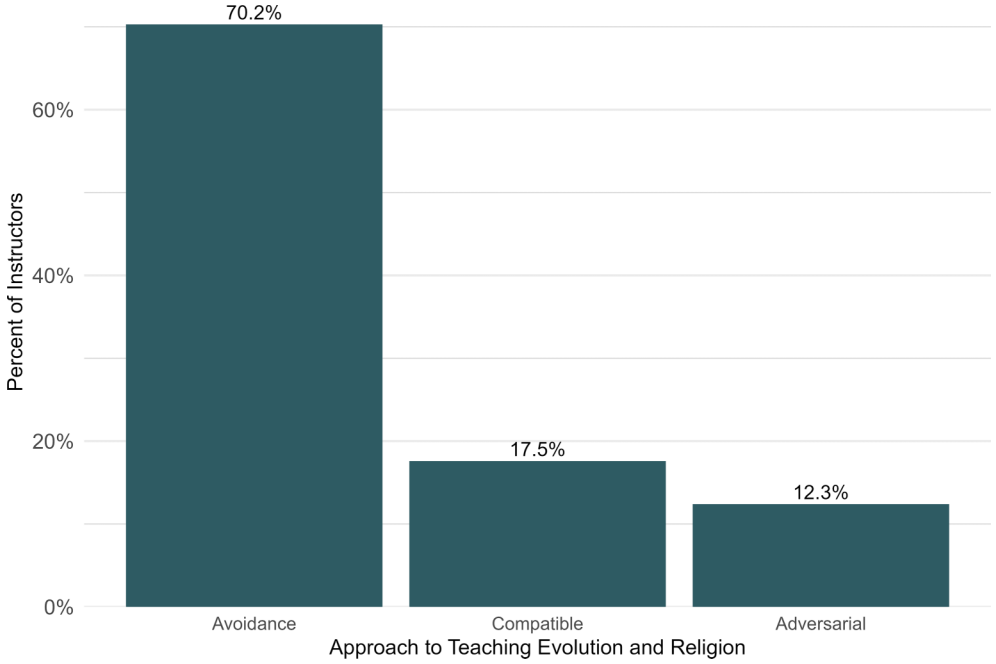


Figure 4. Instructor approaches to discussing evolution and religion in their courses. For instructors who used different approaches depending on the course, they were categorized into compatible or adversarial if they ever used one of those approaches and also sometimes did not. Instructors who reported teaching both adversarial and compatible approaches were classified as adversarial because even a small amount of adversarial instruction has the potential to negatively affect students. No instructors described teaching a completely adversarial relationship between religion and evolution in one course and showing compatibility in another ($n = 171$).

Discussion

This is the first study to our knowledge to explore on a national level, aspects of college evolution instruction that might be a barrier to evolution being used by students as a core concept in their thinking of biology. Instructor treatment of evolution, their instructional goals, their beliefs about the nature of science, and their reluctance to address religion while teaching evolution may be contributing factors.

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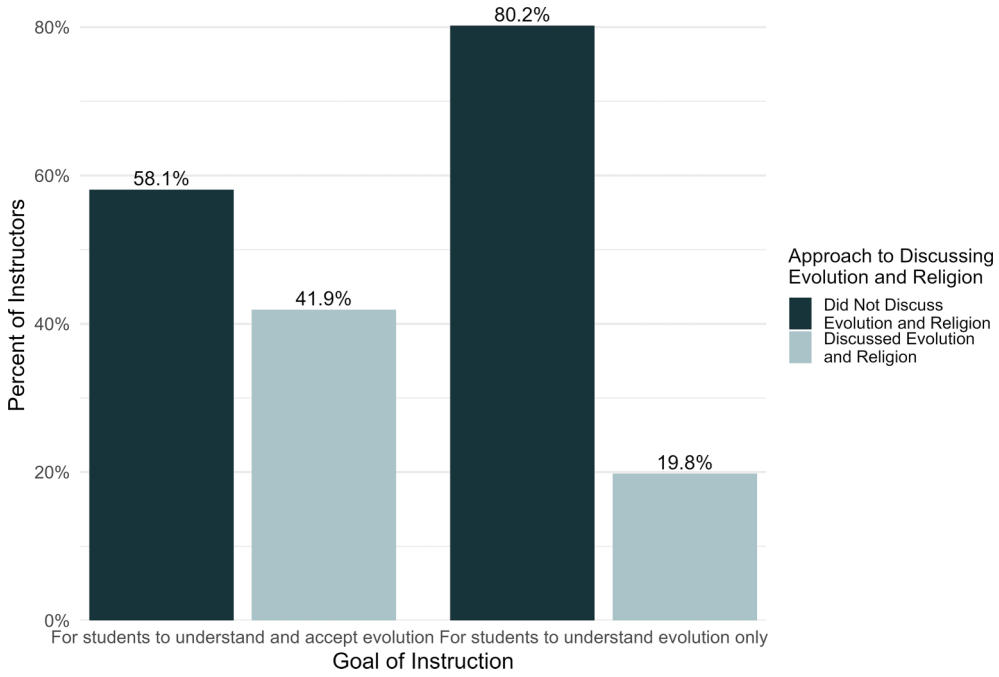


Figure 5. Bar chart presenting evolution instructors' goals for teaching evolution by their approach to religion and evolution in their courses. Instructors with a goal of understanding and acceptance were more likely to discuss the topic ($n = 171$, $x^2 = 7.67$, $p = .0056$).

A total of 88% of instructors think they are using evolution as a core concept to guide their teaching, but only 33% of faculty are doing so in all the biology content courses they teach, and only half of faculty reported weaving evolution throughout at least one of their courses, which highlights a disconnect in instructor perceptions of what they are doing as far as teaching evolution as a core concept. *Vision and Change* specifically discusses that the five core concepts, including evolution, be integrated throughout the curriculum in all classes. Further, nearly a quarter of instructors reported teaching at least one biology content course in which they did not introduce the concept of evolution at all. Although some resources already exist (Brownell et al. 2014, Branchaw et al. 2020), instructors may need more awareness, guidance, and resources about how to integrate the core concepts into their teaching.

We found that instructors were split on whether student acceptance of evolution was a goal of their instruction. As biology educators, it is our job to help students become scientifically literate and be able to accurately evaluate the strength and weakness of evidence for a scientific claim, but half of instructors do not think it is their goal to help students accept evolution, for which some of the strongest scientific evidence exists (Dobzhansky 1973, NAS 2008). Instructors may think aiming for acceptance is authoritarian or would be forcing students to accept evolution, but there are ways to aim for student acceptance of evolution while still fostering student autonomy over their beliefs. First, instructors do not need to include acceptance of evolution as a learning goal on their syllabus or assert to students that they "must accept evolution." In fact, our research shows trying to force students to accept evolution can actually lead to students accepting evolution less (Aini et al. 2024b). However, just as instructors may aim to foster awe in science without requiring it from students, instructors can also aim for acceptance. We can be transparent that we want students to accept evolution and incorporate evolution into their biological thinking because of the strength of scientific evidence, while emphasizing that students are not required to accept evolution.

One barrier to student acceptance of evolution may be that some instructors equate it with atheism. Nearly half of faculty in our study reported that if students believe God/god(s) created the process of evolution, they do not fully accept evolution—an atheistic definition also found among undergraduates (Barnes et al. 2020a). This contrasts with the bounded nature of science, which is agnostic about forces outside the natural world (Huxley and Huxley 1900, Dobzhansky 1973, Collins 2007, Miller 2007; Gould 2011). If instructors present evolution as atheistic, they risk reinforcing misconceptions about science and alienating students in a largely religious society (Pew Research Center 2025). Such views may stem from the idea that belief in a higher power contradicts the nonteleological nature of evolution (Jacob 1977, Mayr 1988). Yet science cannot speak to whether a creator initiated evolution or knew its outcomes (Ruse 2004, Scott 2005). As Dobzhansky (1973) wrote, "It is wrong to hold creation and evolution as mutually exclusive alternatives… I am a creationist and an evolutionist. Evolution is God's, or Nature's, method of creation." Our author team includes both atheists and Christians (see supplemental material for positionality statements), but we agree that personal beliefs lie outside evolution's domain. Students should leave biology courses understanding that all life shares common ancestry and diversified through natural processes; whether a creator is behind this is beyond science.

Only 18% of instructors in our study reported discussing religion when teaching evolution in a way that emphasizes compatibility. This is a lost opportunity, as there is an abundance of research indicating that doing so can lead to an increase in evolution acceptance (Barnes and Brownell 2026) . From qualitative interviews, single classroom studies, quasi-experimental studies, and most recently, a randomized controlled study, the evidence is clear that presenting potential compatibility between evolution and religion reduces students' conflict and increases both their understanding and acceptance of evolution

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(Scharmann 2005, Verhey 2005, Ingram and Nelson 2006, Martin-Hansen 2006, Nehm and Schonfeld 2007, Donnelly et al. 2008, Ladine 2009, Wiles and Alters 2011, Winslow et al. 2011, Southerland and Scharmann 2013, Wiles 2014, Barnes et al. 2017a, Holt et al. 2018, Truong et al. 2018, Bertka et al. 2019, Lindsay et al. 2019, Alkaher et al. 2020, Tolman et al. 2020, 2020b, Ferguson and Jensen 2021, Green and Delgado 2021, Bowen et al. 2022, Ferguson et al. 2022, Kaloi et al. 2022, Aini et al. 2024b, Saefi et al. 2024, Bland and Scoles 2025). Despite this large body of robust evidence for conflict-reducing practices, almost three-quarters of instructors reported that they do not discuss the relationship between religion and evolution when teaching evolution, and although instructors in prior studies have reported that they omit discussions of religion in an attempt to maintain neutrality (Barnes and Brownell 2016, Edwards et al. 2025c), students have reported that an instructor's decision to avoid discussions of religion can be perceived by students as the instructor subscribing to conflict narratives surrounding religion and evolution (Barnes et al. 2017b, Edwards et al. 2024). Further, approximately 8% of faculty in this study admitted to presenting religion as adversarial to evolution. For instructors who do see evolution acceptance as a goal, discussing religion in a way that emphasizes potential compatibility with evolution can be effective. We have found that as little as six minutes of discussion on potential compatibility can increase evolution acceptance (Truong et al. 2018) and that this can even be assigned as a homework video if instructors prefer not to address it in class (Aini et al. 2025). Further, we would argue that even when acceptance is not a personal goal for instructors, due to the potential implications of evolution rejection, incorporating mention of religion in a way that emphasizes compatibility with evolution is important for *all* instructors.

Why were women more likely than men to report an agnostic definition of evolution acceptance? Although women are more likely to be religiously affiliated than men (Pew Research Center 2016), we controlled for religious affiliation and religious history in our regression, indicating that these factors are not driving the observed gender differences. Women also reported being less likely than men to teach for acceptance, so it may be that women are more hesitant to "push" ideas on students, which is in line with research showing that women are less forceful in conflict resolution than men, especially in individualistic societies like the United States (Holt and DeVore 2005). We also know that women are more likely than men to reveal their own stigmatized identities in small group and one-on-one conversations with students (Busch et al. 2023). So, women might be more aware of student attempts at reconciling their religion and evolution and more cognizant about how identities can affect students in class (Henning et al. 2019, Cooper et al. 2020, Busch et al. 2022, Edwards et al. 2024, 2025a, 2025b), which could drive their adoption of more agnostic definitions of evolution acceptance.

We were equally interested in our finding about the *lack* of differences based on religious affiliation and religious history. Existing literature suggests that nonreligious individuals perceive more conflict between religion and evolution than religious individuals (Waite and Spencer 2022); further, for some nonreligious individuals who were religious previously, a perception that religion and evolution cannot coexist could contribute to their decision to ultimately become nonreligious. Thus, we predicted that nonreligious faculty might be more likely to have an atheistic definition of evolution acceptance and avoid religion in their teaching, but we did not see this difference based on reported backgrounds of nonreligious instructors. This implies that there is a greater level of complexity at play than instructors' own religious affiliation when it comes to their decisions to address religion and their definition of evolution acceptance. Nonreligious instructors who were previously religious might be an interesting population to explore in the future to understand their reasoning and decision-making as they currently do not have religious beliefs but may have a greater understanding of religious culture and beliefs than instructors who were never religious. Overall, more research is needed to understand why instructors avoid or choose to discuss religion, have atheistic or agnostic definitions of acceptance, and why they choose understanding versus acceptance as a goal of their instruction, and why they approach teaching evolution as a core concept of biology or not.

Limitations

Self-selection bias is inherent in any survey study; although we implemented strategies to reduce bias by not advertising certain components of the survey (including *Vision and Change*, evolution, or religion) in the email invitation, it could be possible that individuals with less familiarity with *Vision and Change* decided to stop completing the survey when they reached that question or that individuals with certain identities felt uncomfortable answering survey questions. However, given the challenges associated with recruiting a national population of biology instructors, our sample size was sufficient to detect small effect sizes.

Conclusion

Among a national sample of college biology instructors in the United States, we found that evolution is not consistently being incorporated as a core concept of biology that students should learn, accept, and incorporate within their framework for biology and their thinking about the diversity of life. If we want to see evolution acceptance rise and have students apply it as a framework for their thinking about biology, we may need to consider making acceptance of evolution a common goal of our instruction and providing instructors with using evidence-based methods to do so, including weaving evolution throughout all our courses and highlighting potential compatibility between religion and evolution.

Supplemental materials

Supplemental data are available at *BIOSCI* online.

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Author contributions

Erin Grace Rowland-Schaefer (Conceptualization, Data curation, Formal Analysis, Visualization, Writing—original draft, Writing—review & editing), Baylee A Edwards (Conceptualization, Data curation, Writing—original draft, Writing—review & editing), Alison Crowe (Conceptual-

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ization, Writing—review & editing), Elizabeth Barnes (Conceptualization, Data curation, Formal Analysis, Funding acquisition, Investigation, Methodology, Project administration, Writing—original draft, Writing—review & editing), Sara E. Brownell (Conceptualization, Data curation, Formal Analysis, Investigation, Methodology, Project administration, Writing—original draft, Writing—review & editing).

Conflict of interest

The authors declare no conflicts of interest.

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